

Loops in Bash

Definition:

A loop in Bash is a control structure that executes a block of commands multiple times until a certain condition is met or for each item in a list.

Types of Loops in Bash

1. for Loop

Repeats a block of commands for a fixed number of times or items.

Syntax:

```
for variable in list
do
    commands
done
```

Example:

```
for fruit in apple banana cherry
do
    echo "I like $fruit"
done
```

C-style for loop:

```
for (( i=1; i<=5; i++ ))
do
    echo "Number $i"
done
```

2. while Loop

Executes commands as long as a condition is true.

Syntax:

```
while [ condition ]  
do  
    commands  
done
```

Example:

```
count=1  
while [ $count -le 5 ]  
do  
    echo "Count = $count"  
    ((count++))  
done
```

3. until Loop

Executes commands until the condition becomes true (opposite of while).

Syntax:

```
until [ condition ]  
do  
    commands  
done
```

Example:

```
num=1  
until [ $num -gt 5 ]  
do  
    echo "Number = $num"  
    ((num++))  
done
```

done

4. select Loop

Used to create simple menus for user input.

Syntax:

```
select variable in list
```

```
do
```

```
    commands
```

```
done
```

Example:

```
select drink in Coffee Tea Juice
```

```
do
```

```
    echo "You selected $drink"
```

```
    break
```

```
done
```

Loop Control Statements

break - Exit the loop immediately

continue - Skip current iteration

exit - Exit the script

Example:

```
for i in {1..5}
```

```
do
```

```
    if [ $i -eq 3 ]; then
```

```
        continue
```

```
    fi
```

```
    if [ $i -eq 5 ]; then
```

```
break
fi
echo "Number: $i"
done
```

Summary Table

Type	Purpose	Condition Check	Example	
-----	-----	-----	-----	
for	Iterates over items	Automatic	for i in 1 2 3; do ...; done	
while	Loops while true	At start	while [\$x -lt 5]; do ...	
until	Loops until true	At start	until [\$x -gt 5]; do ...	
select	Menu-based loop	User input	select var in list; do ...	

Nested Loop Example

```
for (( i=1; i<=3; i++ ))
do
    echo "Outer loop $i"
    j=1
    while [ $j -le 2 ]
    do
        echo " Inner loop $j"
        ((j++))
    done
done
done
```